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THE EXTENT OF GREATER INDIA, III. PALAEOMAGNETIC DATA FROM THE TIBETAN SEDIMENTARY SERIES, THAKKHOLA REGION, NEPAL HIMALAYA

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The palaeomagnetism of 750 carbonate and sandstone samples from the Tibetan Sedimentary Series (TSS) of the Thakkhola region (north central Nepal) has been studied. This region forms part of India's former leading edge, underthrust by India along the Main Central Thrust (MCT). Samples of Devono-Carboniferous to Early Cretaceous age were collected from 14 localities. Five magnetic components have been distinguished: (1) a predominant component of recent origin, (2) a secondary component probably acquired during early Tertiary collision of India with Asia or offshore island arcs, (3) primary magnetic components of Middle? Permian to Early Cretaceous age, and another two secondary components either tentatively related with the predominant *b*-axes pattern of the TSS (4), or attributed to MCT-related metamorphism of the lower TSS (5).

Declinations of the secondary collision component and the primary components mismatch declination data from the Indian subcontinent by 10–15°. A testable model is proposed, which relates this mismatch to rotational underthrusting of India along the MCT beneath its former leading edge and beneath the Tibetan Plateau. The model accounts for underthrusting of continental lithosphere over 200–350 km at the longitude of the sampled region in central Nepal.

A minimum estimate for the northern extent of India (post-Neotethys formation) is proposed, based on this magnitude of rotational underthrusting and on generally accepted values for crustal shortening across the Himalaya. Within the modified Smith and Hallam reconstruction of Gondwanaland, this estimate satisfies constraints imposed by seafloor spreading data from the eastern Indian Ocean and facies analyses of the western Australian continental margin.

The palaeomagnetic results herein presented as well as those obtained elsewhere in the Himalayan region can be interpreted in support of Molnar and Tapponnier's model for deformation of southern Asia.

1. Introduction

Geological and geophysical studies increasingly focus on tectogenetic interpretations of the Tibetan Plateau and its surrounding mountain chains in terms of evolution of the India-Asia collision. A proper estimate for the northern extent of "Greater" India ([1–3], post-Neotethys formation) and less importantly so for "Gondwanic" India (pre-Neotethys formation) is a prerequisite for further modelling of the continental collision aspects of the plate tectonic theory. Some newly available palaeomagnetic data from southern Tibet, north of the Indus-Tsangpo suture (ITS) zone, have been interpreted [4] in terms of an Eurasian affinity. This interpretation conflicts with geological, palaeontological and palaeobotanical indications that parts of southern Asia north of the ITS zone are original (pre-Neotethys) Gondwanaland fragments [5–21]. Further palaeomagnetic data are needed to settle this controversy. In this paper we confine ourselves to constrain the northern extent of "Greater" India (post-Neotethys, called India from here on) by means of palaeomagnetic studies in accessible parts of the Himalayan region.

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There are currently three prevalent models which describe the evolution of the Tibetan Plateau and its notable twice average crustal thickness in terms of widely varying estimates for the northern extent of India: (a) Dewey and Burke's [22] basement reactivation model, (b) Molnar and Tapponnier's [23,24] lateral displacement model, and (c) Powell and Conaghan's [2,3,25] large-scale underthrusting model.

These models can be constrained on the basis of palaeomagnetic data from the Indian subcontinent [26] and Eurasia [27], together with recently obtained [28–30] and here presented palaeomagnetic data from the Alpine-Himalayan belt. In accordance with the geological record these palaeomagnetic data suggest that initial contact between India and Asia or offshore island arcs was established in about late Palaeocene, and show that this contact occurred at equatorial to low northern latitudes. Seafloor spreading data [3,25,31] and palaeomagnetic data [30] concurrently demonstrate that subsequent convergence of India with Asia exceeded 2500 km. Each of the three above models (a, b and c) individually accounts for this massive convergence only in part. The models are, however, not necessarily mutually exclusive.

In the following we discuss new palaeomagnetic results from the Tibetan Sedimentary Series (TSS, north of the Main Central Thrust) and interpret these results in terms of the magnitude of India's underthrust of the Tibetan Plateau (see also Bingham and Klootwijk [32]). We propose a minimum estimate for India's northern extent in the sense of model (c), and will place indirect constraints on models (a) and (b).

Testing of underthrusting models is facilitated by the unique Himalayan condition that India's former leading edge is preserved at the surface north of the Main Central Thrust (MCT) as the Tibetan Slab and overlying TSS. Comparison of palaeomagnetic data from this former leading edge with data from the Indian subcontinent south of the MCT facilitates determination of "rotational" and northwards "translational" components of underthrusting. A rather precise estimate of the "rotational" component can

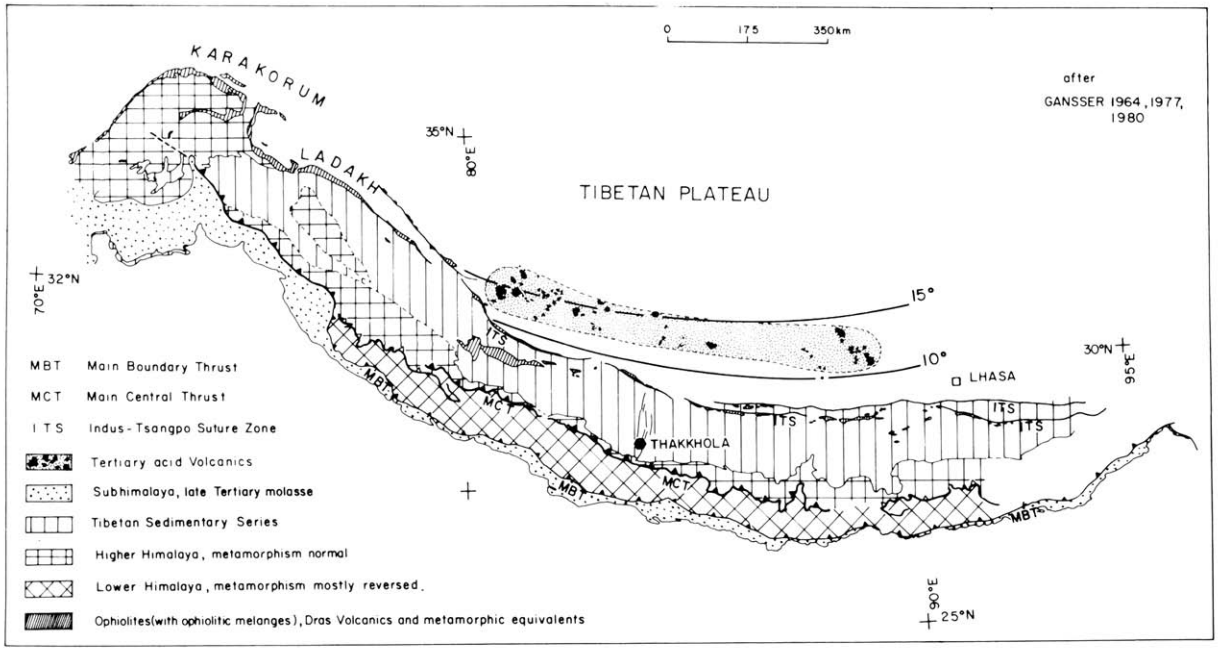


Fig. 1. Main structural features of the Himalaya after Gansser [82–84]. The dashed line indicates the westwards subsurface continuation of the Main Boundary Thrust–Main Central Thrust (MBT-MCT) zone according to seismic observations [75]. The curved lines labeled "10°" and "15°" are estimated surface projections of the frontal edge of India, underthrust along the MCT beneath its former leading edge and beneath the Tibetan Plateau (see text).

be obtained from comparison of declination patterns. The magnitude of the "translational" component may be estimated from comparison of palaeolatitude observations, but with lower precision than for the "rotational" component.

Seafloor spreading [25,33–35] and geological data [36] show that post-collisional convergence of India with Asia was characterized by an appreciable counterclockwise rotational movement of India with respect to Asia. This may have led to more pronounced underthrusting along the MCT to the east [37], and an eastwards increase in differential movement between India and its former leading edge may be surmised.

As representative for this leading edge we selected the TSS of the Thakkhola region (north central Nepal), situated in the central Himalaya (Fig. 1). This region is relatively easily accessible, has been studied in considerable detail [38–40] and has undergone a lower degree of metamorphism than adjacent regions of the Tibetan Himalaya.

2. Regional geological setting

Convergence and collision of India with Asia took place at first along the ITS (Fig. 1). In the west, India collided initially with an island arc off south central Asia [41,42]. The eastwards extension of this island arc has not been established as yet. A further eastwards extension has been suggested [25,43–46], but the Transhimalayan plutonic belt north of the ITS (Kangdese belt) also has been interpreted [47] as the result of Andean-type subduction. The collision process – i.e. deformation of non-matching juxtaposed continental outlines, closure of a presumed marginal basin along the Shyok zone north of the ITS [41], subduction of continental lithosphere along the ITS, and finally immobilization of the same – started not later than Late Palaeocene and may have continued till Late Eocene. Timing and evolution of the collision process can be determined from the geological record of the opposing continental margins (see summaries in references 25, 30, 48, 49), and is well reflected in changes in plate movements [3,23,25,35] and in palaeomagnetic data from DSDP cores of the Indian plate [26,50] and from the ITS in the Ladakh region [16]. After immobilization of the ITS at about Late Eocene, a new zone of weakness (the future MCT, Fig. 1) started to develop within the northern margin of India about 100–200 km south of the ITS. Large-scale intracontinental underthrusting along the MCT occurred during the Miocene [2,3], but started possibly already during the Late Oligocene [25,49], or maybe earlier (P. Le Fort, personal communication, 1980). The MCT is structurally very pronounced in the central Himalaya, but merges with the Main Boundary Thrust (MBT) in the western Himalaya (Fig. 1). Its overriding (northern) slab is characterized by a basal gneissic zone (Tibetan slab, Central Crystalline Axis) which shows a pronounced inverted metamorphism. The normally overlying TSS display an upwards decreasing metamorphism.

The nature of the MBT (Fig. 1) has been variously interpreted. Powell and Conaghan [2] interpret the MBT as the frontal thrust of a telescoped sedimentary cover, sheared off its basement due to continuing underthrusting along the MCT. In contrast, Le Fort [49] postulates that intracontinental underthrusting shifted from the MCT to the MBT at about 10 m.y. ago.

3. Geological setting of the sampled region

The Thakkhola graben (Fig. 2) in north central Nepal, north of the MCT, is bounded by NNE-SSW-oriented postorogenic (Plio-Pleistocene, or possibly Miocene) faults transverse to the main Himalayan structural trend. Downthrow along the western boundary fault reaches about 3000 m [39], but along the eastern boundary fault is very much less. Within the graben MCT-related metamorphism has visibly affected lower Palaeozoic rocks only, but extends up to Triassic rocks outside the graben.

A lower Palaeozoic to Lower Cretaceous succession is well exposed from the north face of the Annapurna-Nilgiri massif northwards into the Thakkhola graben. This succession is stratigraphically and structurally continuable westwards into the Dolpo region [51–54] and eastwards into the Nyi-Shang (Manang) region [55], and has been well described [38–40,54, 56–58]. Only broad outlines of the regional geology, of interest for the present study, will be sketched here.

The Palaeozoic/Mesozoic sedimentary succession

has a total thickness of more than 10,000 m, deposited in a relatively quiet basin that underwent only epeirogenic movements as reflected in several distinct discontinuities. Palaeozoic sedimentation was mainly epicontinental with a more pelagic Siluro-Devonian interlude. The lower Palaeozoic is developed mainly in calcareous facies. A sandy facies appears already in the Lower Ordovician and is predominant in the Permian regressive succession, succeeded by a Triassic transgressive facies. Mesozoic sedimentation took place in subsiding shallow basins and is characterized by detrital influx. The Triassic and the Lower and Middle Jurassic are largely developed in carbonate facies, with sandy intercalations particularly in the Rhaetian and less predominantly in the Lower Triassic and Middle Jurassic. Upper Jurassic sedimentation under stagnating conditions was followed by Lower Cretaceous continental-type sedimentation, with sudden appearance of volcanic detritus in Lower Aptian rocks ([39,40], M. Colchen, personal communication, 1980).

The lower structural element, the Tibetan Slab, dips monoclinally north. The overlying TSS is unroofed more deeply southwards with lower Palaeozoic rocks beautifully exposed in the Nilgiri-Annapurna massif. Fold axes within the TSS of the Thakkhola graben and of adjacent regions show a minor but consistent westwards plunge [59]. The TSS is characterized structurally by a double vergent setting of folds and thrusts. These are well continuable along strike, with the thrusts being more pronounced to the east of the graben [55]. Four phases of deformation may have occurred between the Middle Eocene and Pleistocene [56]. The earlier three deformation phases are characterized by WNW-ESE-oriented *b*-axes, with the final phase resulting in large submeridional oriented domes.

4. Sampling

More than 700 cores were collected with a portable drill. Enthusiastic watercarrying by our Nepali porters enabled drilling up to 4000 m (Fig. 2, locations 6, 7). An additional 40 blocksamples were collected from a quartzitic sandstone succession (Fig. 2, location 13). Samples were collected as stratigraphic successions, generally one sample per bed. All samples

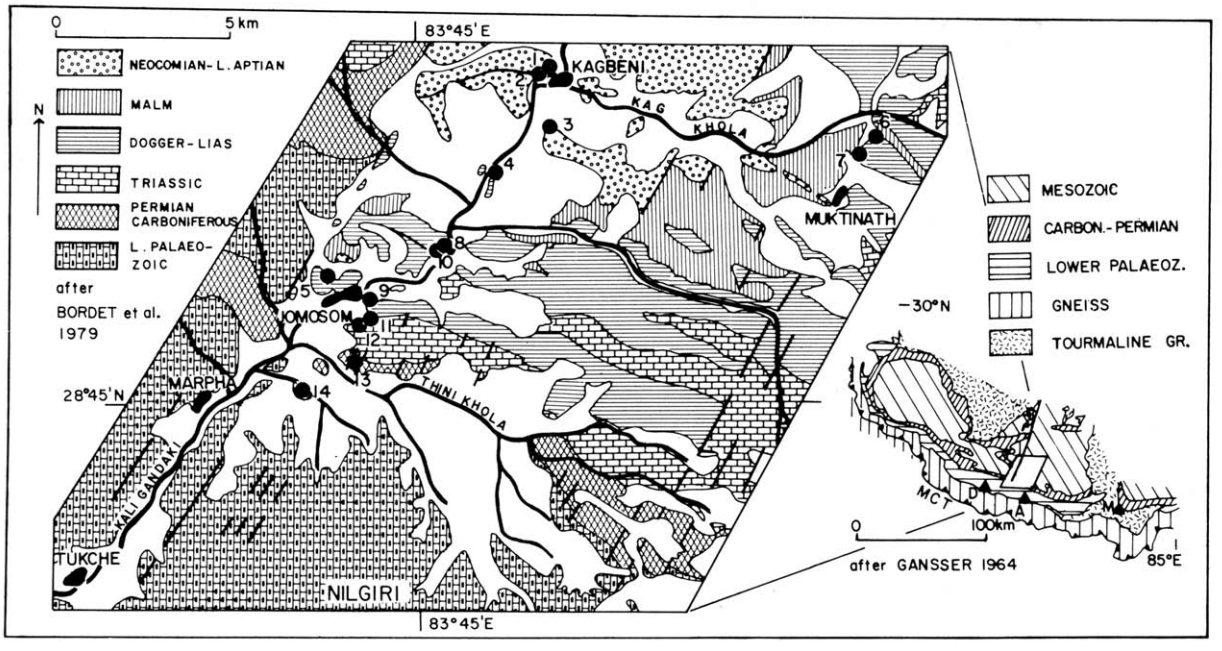


Fig. 2. Geology of the sampled region after Bordet et al. [40]. Sample locations are indicated by full dots. See text for description. Inset after Gansser [84]: D = Dhaulagiri, A = Annapurna, M = Manaslu.

were oriented in the field with a magnetic and a sun compass, using an apparatus after Embleton and Edwards [60]. Bedding data are given in Table 2. Section lengths are given as stratigraphic thicknesses.

5. Localities

See Fig. 2 and Table 2.

Tilicho Pass Formation–Ice Lake Formation (location 14). Fifty-one cores (NSCD1-51) were collected from a northward overturned succession exposed in the south bank of the Shokang Khola, opposite the ruins of Shokang (28°45′N, 83°44′E; altitude 2900 m). This succession straddles the Devono-Carboniferous boundary [40, p. 98]. Twenty-five cores (NSCD1-25) were taken over 15 m in a tectonically lower succession of thin bedded to massive dark grey limestones of Early Carboniferous age. Another 26 cores (NSCD26-51) were collected over 27 m through a tectonically upper succession of sandstones, sandy limestones and dolomitic limestones of Late Devonian age. An intervening stratigraphic interval of about 20 m has not been sampled.

Thini Chu Formation (location 13). Forty block-samples (NTPA1-40) were collected over 20 m through a pronounced massive quartzitic sandstone succession (28°45.5′N, 83°44.5′E; altitude 3240 m [40,p. 105]) exposed below the main sandstone escarpment which runs southeast of Thinigaon. The age of the sampled rock unit is not well defined within the Permo-Carboniferous. We have followed Waterhouse's [54] assignment of this succession to the Middle? Permian Shokang member of the Panjang Formation.

Thinigaon Limestone (locations 12, 11). Fifty-seven cores (NTTR1-57) were collected through two successions (location 12) to the east of the Thinigaon-Nupra footpath up the Thini ridge [40, pp. 129–131]; eighteen cores (NTTR1-18) were collected over 30 m in steeply northwards overturned slightly sandy limestones (altitude 3020 m), and another 39 cores (NTTR19-57) were collected over 18 m in northwards dipping blue-grey thin bedded limestones exposed somewhat higher up the ridge (altitude 3060 m). Both successions are of Ladinian/Carnian age [40]. In another succession still higher up along the same footpath (location 11, altitude 3130 m) we sampled 76 cores (NJUT1-76) over 90 m in slightly sandy limestones of Norian age.

Jomosom Quartzite (location 10). Fifty-five cores (NJRH1-55) were collected from sandstones, sandy and marly limestones and Lumachelle beds over 56 m in a uniformly northwards dipping succession of Rhaetian age, exposed along the west bank of the Kali Gandaki about 4 km north of Jomosom. Samples were collected up to the base of the overlying Jomosom Limestone.

Jomosom Limestone (locations 9, 8, 7). This formation is made up of thick bedded dark grey platform-type limestones and forms conspicuous ridges in the terrain. The beds are largely devoid of fossils, but some recent microfaunal findings [57] confirm the Liassic age inferred from their stratigraphic position. Samples were collected at three localities. Fifty-two cores (NJJL1-52, location 9) were drilled over 47 m in slightly eastwards dipping beds exposed near the core of the northward overturned synclinal structure which dominates the ridge directly east of Jomosom (28°46′N, 83°44.5′E). Another 82 cores (NJLI1-82, location 8) were collected over 75 m through the basal Jomosom Limestone directly above the sampled Jomosom Quartzite succession of locality 10. Finally 55 cores (NCJL1-55, location 7) were collected over 40 m in the scarp of the eastern graben fault, directly east of Chhokor (28°48′N, 83°50′E, altitude 3850 m).

Lumachelle Formation (locations, 6, 5, 4). The basal part of this Middle Jurassic succession has been sampled in the fault scarp opposite Chhokor (location 6, altitude 4000 m) above the Jomosom Limestone of locality 7. Forty-seven cores (NCDL1-47) were collected over 13 m in thin bedded to massive blue grey limestones. Another 38 cores (NJDO1-38) were taken in the southern limb of a tight synclinal structure directly west of Jomosom (location 5 [40, pp. 147–149]). Twenty-five cores (NJDO1-25) were taken through a 9-m-thick upper succession made up of sandy and ferruginous limestones and lumachelle beds of early Callovian age. Thirteen more cores (NJD025-38) were taken, stratigraphically about

70 m lower, through a 13-m-thick succession of thick bedded limestones and lumachelle beds of Bajocian age. On the east bank of the Kali Gandaki directly opposite Bagung (location 4, 28°47'N, 83°46'E) another 43 cores (NDBA1-43) were collected over 24 m in thick bedded dark blue limestones [40, p. 151] of Bajocian/Bathonian age.

Kagbeni Sandstone (locations 3, 2). Continental-type massive quartzitic and arkosic sandstones of Neocomian to Barremian age were sampled in a steeply northwards dipping succession about 2 km south of Kagbeni in the east bank of the Kali Gandaki (location 3), and also at a locality directly opposite Kagbeni (28°49'N, 83°46.5'E) on the west bank of the Kali Gandaki (location 2). Forty cores (NKGW1-40) were taken at locality 3 over 50 m and 34 cores (NKWE1-34) were taken at locality 2 through a 60-m-thick succession reaching up to the base of the overlying Dzong Sandstone.

Dzong Sandstone (location 1). An extensive succession of Dzong Sandstone was sampled opposite Kagbeni, directly above the Kagbeni Sandstone sampled at locality 2. Ninety-five cores (NKGV1-95) were taken over 150 m in glauconitic sandstones and dark siltstones. This formation is well dated as Early Aptian. Its higher part is characterized by sudden appearance of volcanic detritus [39,40].

6. Method of analysis

Prior to slicing all samples were washed in dilute hydrochloric acid to remove magnetic stains possibly resulting from the field drilling process. In general at least two specimens of 2.5 cm diameter and 2.2 cm length were sliced from each core. One specimen from every third or fourth successive sample was subjected to progressive thermal demagnetization from room temperature up to 560°C or 675°C. Remanence measurements were taken in 15–20 steps until the remanence had become too weak for meaningful measurement on our cryogenic magnetometer (below 3×10^{-3} mA/m) or until excessive build up of a viscous remanent magnetization (VRM), apparent after heating at higher temperatures, prevented further remanence measurements. The bulk of the remaining specimens was subjected to partial progressive thermal demagnetization in at least 10 steps over a selected temperature interval, generally reaching from 150 to 520°C for the carbonates and up to 560°C for the sandstones. All heatings were done in furnaces after McElhinny et al. [61], with the ambient field maintained below 5 nT over the 2-hour heating-cooling cycle. All specimens were flushed with argon gas prior to the thermal demagnetization treatment and all heatings were done in argon atmosphere to reduce formation of new magnetic phases. At all other times, the specimens were stored magnetically shielded in mu-metal boxes (ambient field less than 50 nT) to avoid excessive build up of VRM.

Changes in magnetic susceptibility over the course of the progressive demagnetization treatment were monitored with a Digico bulk susceptibility unit. All remanent magnetization measurements were made on a two-axis superconducting rock magnetometer (ScT, noise level less than 2×10^{-2} mA/m), which was interfaced for on-line data reduction with a Digico minicomputer, more recently with a Hewlett Packard 2116B.

The magnetic content of each specimen was analyzed from Zijderveld plots [62,63] and directions of individual components were determined using a principal component analysis program adapted from Kirschvink [64]. Blocking temperature spectra of individual magnetic components commonly overlapped in the carbonates. When such specimens failed to respond satisfactorily to the Zijderveld-type analysis, a subsequent search for cryptic component directions [65] was made following an adapted version of Kirschvink's [64] program. Within-sample variation of magnetic properties and directions was rather pronounced in part of the sample collection. Therefore, unit weight was given throughout the statistical analysis to individual specimen directions, obtained according to either of the above analysis methods, rather than to sample-mean directions.

7. Results and interpretation (see Fig. 5)

Five different magnetic components have been distinguished; their magnetic characteristics are summarized hereunder and in Table 1. Mean directions for each component are given along with statistical

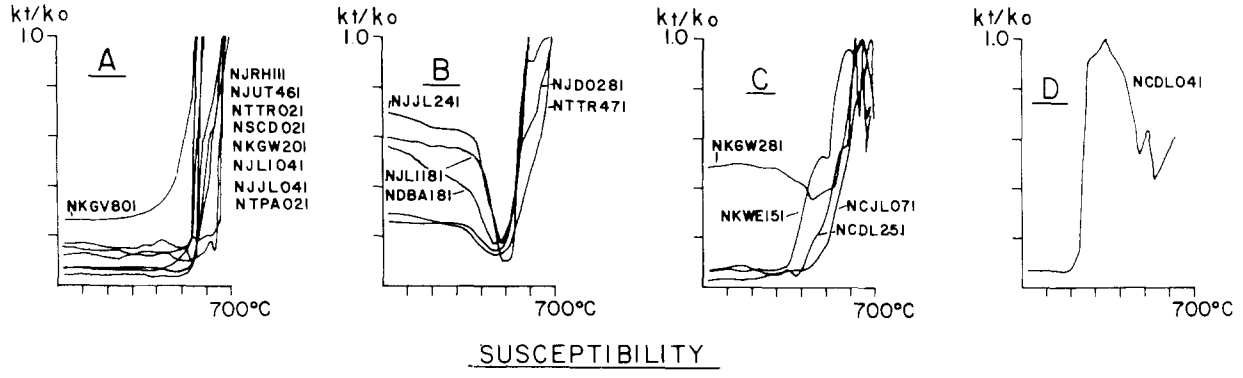


Fig. 3. Representative normalized curves showing changes in magnetic susceptibility after treatment at higher temperatures.

parameters in Table 2. To avoid undue confusion we have named the various components here in accordance with their interpretation, which follows thereafter.

(A) A predominant recent field component of normal and less frequently reversed polarity which became gradually eliminated below 270°C, and/or between 300°C and up to 450°C, respectively.

(B) A secondary component of mostly normal polarity, slightly upward and NNW directed (corrected for bedding), which became eliminated between 300 and 470°C. This component is thought to have been acquired during initial collision of India with Asia.

(C) Primary magnetic components, predominantly of normal polarity and ranging in age from Middle? Permian to Early Cretaceous, which became eliminated in general at temperatures from 300 up to 560°C.

(D) A secondary magnetic component attributed to metamorphism associated with the MCT was observed only in the southernmost sampled locality (Fig. 2, location 14). In these samples a very uniform pattern of a normal polarity and a succeeding reversed polarity became eliminated between 200–270°C and 270–330°C, respectively.

(E) A west- to northwest-directed component of low downwards inclination (not corrected for bedding) which became eliminated in general between 300 and 500°C. The origin of this component is not well understood. A relation is surmised with the similarly directed *b*-axes pattern of the TSS.

Susceptibility increased in general at an excessive rate after heating above 500°C (Fig. 3). Remanence measurements after the higher temperature heatings were in general impeded by a concomittant excessive increase in a laboratory-induced and very soft VRM. Samples from the Lower Cretaceous sandstone-siltstone successions (NKGW, NKWE, NKGV) and from some of the carbonate successions, showed a notable but not excessive susceptibility increase already at about 400°C (Fig. 3C). In some of the other carbonate successions, a marked decrease in susceptibility between 400 and 500°C preceded a subsequent increase (Fig. 3B).

The magnetic remanence of the specimens was complex as was to be expected from the tectonized nature of the sampled region. Separation and identification of individual magnetic components was occasionally ambiguous.

Dispersed blocking temperature spectra for the various magnetic components were commonly observed in the carbonates (Fig. 4A, B), with spectra from the various components overlapping within and between specimens. More discrete spectra were observed in general only in the sandstone samples (Fig. 4D), and in those carbonate samples which displayed a normal-reversed sequence of secondary components (Fig. 4C). There was in general no unique relationship between individual components and blocking temperature spectra at temperatures above 300°C. Our sampling method, sectionwise through a variety of

TABLE 1

Summary of magnetic behaviour of magnetic components (blocking temperature spectra)

Formation	Sample location code	Components				
		A	B	C	D	E
		recent field	collision	primary	MCT-related	W-NW
Dzong Sst. [a]	NKGV 1	<270–<350°C not dominant	200–480°C in part of specimens	N polarity <500°C (Fig. 5F)		<500°C in some specimens
Kagbeni Sst. [b]	NKGW 2	<200–<350°C in part of specimens	150–350°C	350–560°C		240–560°C only in top of sampled succession
Kagbeni Sst. [c]	NKWE 3	<200°C not dominant; also 560–670°C in some specimens (Fig. 5B)	150–350°C occasionally reversed polarity	350–560°C		350–560°C in some specimens
Lumachelle Fm. [d]	NDBA 4	N polarity <250–<280°C; R polarity, 280–360°C		<520°C (Fig. 5G)		
Lumachelle Fm.	NJDO 5	N polarity predominant <300°C, but 600–650°C in base of upper succession; R polarity 300–500°C in basal succession only		300–500°C in few specimens only, low precision		R polarity 300–470°C
Lumachelle Fm. [e]	NCDL 6	N polarity <200–<245°C; R polarity 200–440°C		300–520°C in few specimens only, low precision		
Jomosom Lst.	NCJL 7	N polarity predominant <280°C; R polarity 280–400°C in part of specimens		400–480°C N polarity, occasionally R polarity		
Jomosom Lst.	NJLI 8	N polarity <300°C predominant; R polarity 300–400°C, occasionally 500°C, in lower part of succession only	300–500°C (Fig. 5E)	300–500°C		
Jomosom Lst.	NJJL 9	N polarity <300°C (>90% of initial NRM); R polarity 300–450°C (Fig. 5A)		400–470°C in a few specimens only		
Jomosom Qzt.	NJRH 10	<300°C predominant; in part of specimens also 300–560°C (Fig. 6D)	300–560°C	300–560°C		300–560°C in a few specimens from top of succession

TABLE 1 (continued)

Formation	Sample location code	Components				
		A	B	C	D	E
		recent field	collision	primary	MCT-related	W-NW
Thinigaon Lst. [f]	NJUT 11	<270°C predominant (Fig. 6C); in part of specimens also 300–500°C (Fig. 6B)	300–500°C (Fig. 5D)	300–500°C		300–400°C (Fig. 5C, D)
Thinigaon Lst. [g]	NTTR 12	<280°C, occasionally <320°C predominant (Fig. 5H); in part of specimens also 300–500°C		300–500°C (Fig. 5H)		300–500°C
Thini Chu Fm. [h]	NTPA 13	not predominant (Fig. 6C)		300–500°C		300–500°C
Tilicho Pass Fm. [i]–Ice Lake Fm.	NSCD 14	<200°C, about 80% of initial NRM (Fig. 4C)			N polarity 200–270°C R polarity 270–330°C (Fig. 5I)	

[a] Thermally dispersed (Fig. 4A) and thermally discrete (Fig. 4D, 500–560°C) blocking temperature spectra.
[b] Most specimens showed thermally dispersed (Fig. 4A), and only a minority of the specimens showed thermally discrete (Fig. 4D, 500–560°C) blocking temperature spectra.
[c] Thermally dispersed blocking temperature spectra (Fig. 4A). Larger directional scatter than in NKGW succession, probably because of the position of the NKWE succession immediately above a southwards directed thrust (Fig. 2). Some specimens showed an unaccountable NE direction of low inclination between 350 and 560°C.
[d] Very uniform magnetic behaviour.
[e] Large susceptibility increase at 250°C (Fig. 3D).
[f] A few specimens showed a northeastwards and slightly upwards inclined component whose origin is not understood.
[g] Component directions from some of the specimens showed a streaking between the C- and the E-component groupings.
[h] No consistent magnetic behaviour. Large between specimens spread of characteristic directions. In general thermally dispersed blocking temperature spectra. A few specimens showed a discrete blocking temperature range above 600°C (Fig. 4D).
[i] Very uniform magnetic behaviour.

TABLE 2

Summary of palaeomagnetic directions and pole positions

Formation [1]	Age	Initial NRM data	
		N [2]	Intensity [3]
A. Recent field component, normal polarity and lower blocking temperature range			
Dzong Sst. (NKGV, 1)			
Kagbeni Sst. (NKGW, 2)			
Kagbeni Sst. (NKWE, 3)			
Lumachelle Fm. (NDBA, 4)			
Lumachelle Fm. (NJDO, 5)			
Lumachelle Fm. (NCDL, 6)			
Jomosom Lst. (NJJL, 9)			
Jomosom Lst. (NCJL, 7)			
Jomosom Lst. (NJLI, 8)			
Jomosom Qzt. (NJRH, 10)			
Thinigaon Lst. (NJUT, 11)			
Thinigaon Lst. (NTTR, 12)			
Thini Chu Fm. (NTPA, 13)			
Tilicho Pass Fm.–Ice Lake Fm. (NSCD, 14)			
A. Recent field component, reversed polarity and intermediate blocking temperature range			
Lumachelle Fm. (NDBA, 4)			
Lumachelle Fm. (NCDL, 6)			
Jomosom Lst. (NJJL, 9)			
Jomosom Lst. (NCJL, 7)			
Jomosom Lst. (NJLI, 8)			
Thinigaon Lst. (NJUT, 11)			
A. Recent field component, normal polarity and high blocking temperature range			
Jomosom Qzt. (NJRH, 10)			
Thinigaon Lst. (NJUT, 11)			
Thinigaon Lst. (NTTR, 12)			
B. Secondary collison component (50–60 m.y.)			
Dzong Sst. (NKGV, 1)			
Kagbeni Sst. (NKGW, 2)			
Kagbeni Sst. (NKWE, 3)			
Lumachelle Fm. (NJDO, 5)			
Jomosom Lst. (NJLI, 8)			
Jomosom Qzt. (NJRH, 10)			
Thinigaon Lst. (NJUT, 11)			
Thini Chu Fm. (NTPA, 13)			
C. Primary component			
Dzong Sst. (NKGV, 1)	L. Aptian	95 (139)	385 – 6349
Kagbeni Sst. (NKGW, 2)	Wealden	40 (63)	365 – 3918

Mean specimen direction					South pole position				Bedding	
D (°)	I (°)	k	α_{95}	N'	lat. (°S)	long. (°E)	dp (°)	dm (°)	strike (°)	dip (°)
(before bedding correction)										
3	+48	51.5	3	45						
3.5	+43.5	32	7	15						
0	+46	50.5	8.5	7						
0.5	+46.5	37	1	44						
353	+43	95	2	61						
356.5	+43	73	2	60						
356.5	+54.5	254.5	1	57						
5.5	+50.5	50	3	44						
357	+49	57.5	1.5	134						
356.5	+45	75.5	2	85						
0	+60.5	138	1	118						
358	+34.5	50.5	2.5	71						
343	+35	22.5	13	7						
354.5	+57	112.5	1.5	62						
(before bedding correction)										
195.5	−43	18	7	26						
207	−58.5	9	6.5	55						
154.5	−53	18.5	5	49						
197.5	−45.5	39.5	5	20						
160.5	−54.5	25	12.5	7						
189	−62	33.5	21.5	4						
(after bedding correction)										
177	−43.5	18.5	6.5							
187	−29.5	13	5.5							
176	−53.5	43.5	3							
195	+9	35.5	5.5							
172.5	−7.5	25	11.5							
195.5	−44.5	38.5	12.5							
(before bedding correction)										
345.5	+45	40.5	5	20						
0	+60	103	1	23						
4.5	+39.5	23	19.5	4						
(after bedding correction)										
348	−6.5	23.5	3.5	67	−56	106	2	3.5		
2	−12	16	8	22	−55	80	4	8		
347.5	−5	16.5	8.5	18	−56.5	106.5	4.5	8.5		
352	+7	27.5	15	5	−63.5	102	7.5	15		
346	−4	35	2.5	77	−56.5	110	1	2.5		
346	−16	32	4.5	36	−50.5	106	2.5	4.5		
6.5	−9.5	39	6	17	−56	72	3	6		
346	−1.5	29.5	14.5	5	−57.5	110	7	14.5		
(after bedding correction)										
326	−61.5	12	5	78	−12	109	6	7.5	285–300	58– 73N
333.5	−56.5	15.5	6.5	34	−19.5	106	6.5	9.5	280–285	55– 76N

TABLE 2 (continued)

Formation [1]		Age	Initial NRM data	
			N [2]	Intensity [3]
Kagbeni Sst.	(NKWE, 3)	Wealden	34 (47)	576 – 7297
Lumachelle Fm.	(NDBA, 4)	Dogger	43 (49)	1524 – 10,345
Lumachelle Fm.	(NJDO, 5)	Dogger	38 (64)	1599 –153,260
Lumachelle Fm.	(NCDL, 6)	Dogger	47 (64)	2 – 378
Jomosom Lst.	(NJJL, 9)	Lias	42 (57)	1700 – 13,391
Jomosom Lst.	(NCJL, 7)	Lias	55 (54)	7 – 2630
Jomosom Lst.	(NJLI, 8)	Lias	82 (134)	624 – 17,294
Jomosom Qzt.	(NJRH, 10)	Rhaetian	55 (89)	1123 – 27,488
Thinigaon Lst.	(NJUT, 11)	Norian	76 (89)	825 – 22,707
Thinigaon Lst.	(NTTR, 12)	Ladinian-Carnian	57 (73)	639 – 13,117
Thini Chu Fm.	(NTPA, 13)	Middle? Permian	40 (86)	220 – 3345
D. Secondary MCT-related component				
Tilicho Pass FM.– Ice Lake Fm.	(NSCD, 14)	Devono-Carboniferous	51 (66)	86.2– 20,975
As above				

[1] Sample notation and location number according to Fig. 2 are given between parentheses.
[2] Number of specimens subjected to thermal demagnetization is given between parentheses.

lithologies, no doubt contributed heavily to this non-uniqueness. For these reasons, interpretation of individual specimen directions had to be based more on directional grouping within contoured frequency distributions than on grouping according to magnetic properties. Advantages and disadvantages of such a procedure have been discussed elsewhere [66–70].

Mean directions for the normal polarity recent field component (A) group around the axial geomagnetic dipole field direction (inclination = +48°, Fig. 6C, D). The simultaneous occurrence of such components in magnetic domains with both low and high blocking temperature spectra can be explained as a direct consequence of low-temperature oxidation. Mean directions for the reversed polarity recent field component (A) are less well grouped, with inclinations generally slightly steeper than the inclination of the axial geocentric dipole field. This may indicate acquisition prior to the youngest tectonic movement, the more so as these components were observed in

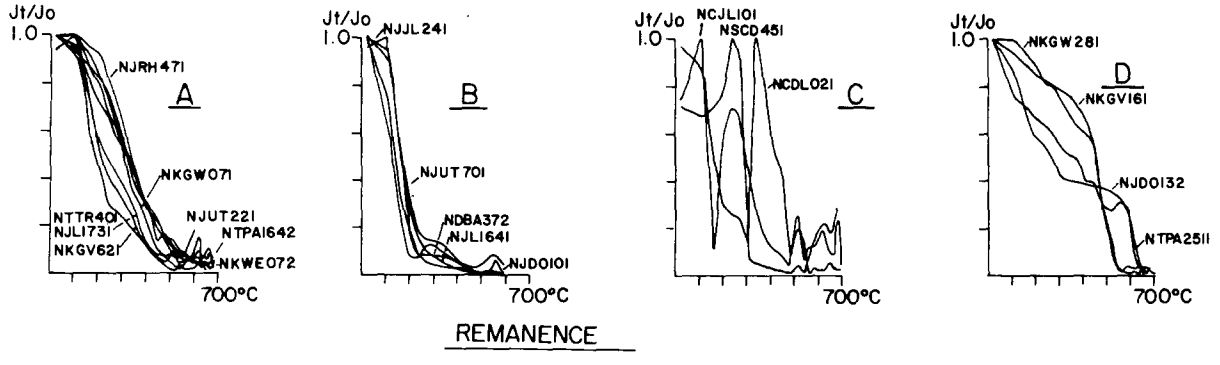


Fig. 4. Representative normalized intensity decay curves of the remanent magnetization during thermal treatment.

Mean specimen direction					South pole position				Bedding	
D (°)	I (°)	k	α_{95}	N'	lat. (°S)	long. (°E)	dp (°)	dm (°)	strike (°)	dip (°)
322.5	−55.5	14	8	24	−16	114.5	8.5	11.5	283–295	65– 70N
308	−61	12	8.5	27	−4.5	119.5	9.5	12.5	130–225	13– 40S
318.5	−45	19	12	9	−22	123.5	9.5	15.5	295–305	62– 65N
315	−63	14	14	9	−6	114	17.5	22.5	248–255	34– 35N
332.5	−55.5	14.5	18	6	−20	107	18.6	26	330–370	12– 22E
307	−59.5	10.5	10	22	−5	121.5	11.5	15	275–283	53– 56N
328.5	−42	14	6	45	−29	116.5	4.5	7.5	280–285	51– 54N
323	−48	11	8	32	−22	118.5	6.8	10.5	280–285	38– 45N
324.5	−43	10	10	22	−26.5	119.5	8	13	110–125	95–105N
329	−47	21	5.5	35	−26	114	4.5	7	270–290	53– 90N
291.5	−63.5	10	7.5	41	−13.5	104	2	2.5	280	42.5N
(before bedding correction)										
309.5	+72.5	23.7	4	60	normal polarity				90– 98	131–152S
78	−66	58	2.5	55	reversed polarity					

[3] Intensity in units of 10^{-3} mA/m.
[4] Number of specimens wherein the component concerned could be isolated.

particular in samples from localities 6 and 7 which are situated in the scarp of the eastern boundary fault of the Thakkhola graben.

Tectonic activity persisted during the Tertiary until very recent time, so the fold test is not particularly informative on the time of acquisition of the various magnetic components. Identification and interpretation of in particular the NNW-directed collision component (B) and the primary components (C) had to be based therefore largely on comparison of their directions with the directional pattern expected for the Thakkhola region according to palaeomagnetic data from the Indian subcontinent (Fig. 9B). Being the sole interpretation method at hand, such a comparison proved to be particularly useful in the present case, because of India's extensive northwards drift and the consequent length of its apparent polar wander path (APWP).

There is a marked improvement in grouping of mean collision component directions after correction for bedding (Fig. 6A). Comparison of the so obtained slightly upwards pointing directions with the Indian palaeomagnetic direction pattern (Fig. 9A, B) indicates that these components were acquired between 50 and 60 m.y. ago. This timing coincides with acquisition of a K-Ar overprint observed by Krummenacher (in [40]) in various regions of Nepal. As discussed above, collision of India with Asia occurred during this period and both the magnetic and the K-Ar overprint may therefore be attributed to regional heating upon collision and subsequent cooling. Similar collision components have been observed elsewhere in northern Indo-Pakistan, i.e. in Baluchistan [29], Ladakh [30] and possibly also in the eastern Hindu Kush [71].

After correction for bedding the primary directions show an improved grouping in the northwest quadrant with moderately steep upwards inclinations (Fig. 6B). Their interformational spread is thought to reflect an early Mesozoic loop in the Indian directional pattern as will be discussed (compare Figs. 6B or 9A with 9B),

Directions observed throughout the sampled region for the west- to northwest-directed component ([E], not corrected for bedding) are shown in a contoured frequency distribution in Fig. 7. Their distribution resembles that of the b-2 axes observed in the TSS [72, p. 308], but with an additional streaking towards the present field direction (Fig. 7B). This component possibly represents preferential growth of new magnetic phases within the dominant cleavage plane under influence of the ambient field.

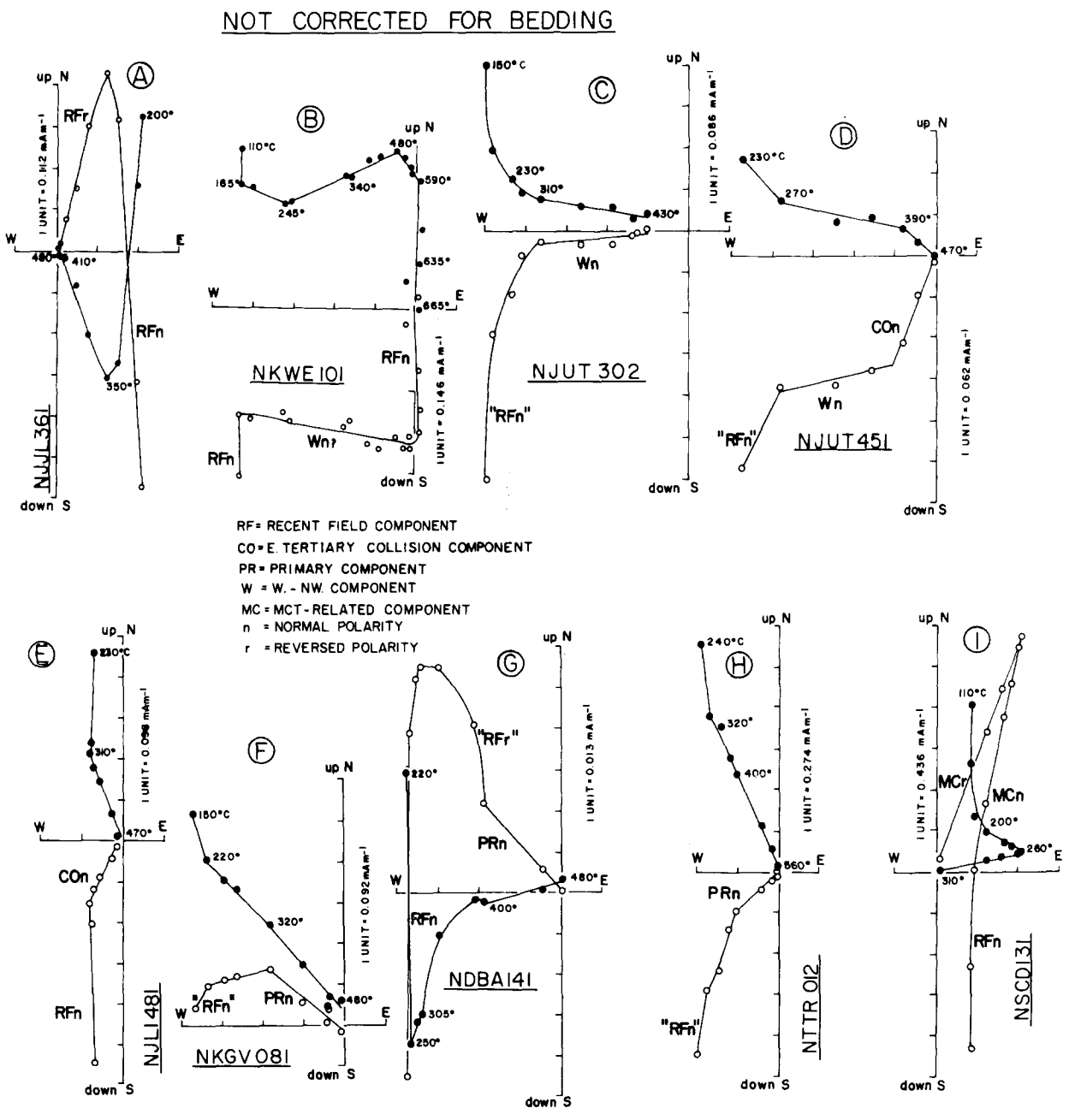


Fig. 5. Demagnetization diagrams of some representative specimens. The points denote successive positions – in orthogonal projection – of the end points of the resultant magnetization vector during progressive thermal demagnetization. Open circles denote projections on the vertical east-west plane, dots denote projections on the horizontal plane. Numbers denote successive peak temperature values. Component references between quotations indicate predominant presence, but incomplete separation of such components.

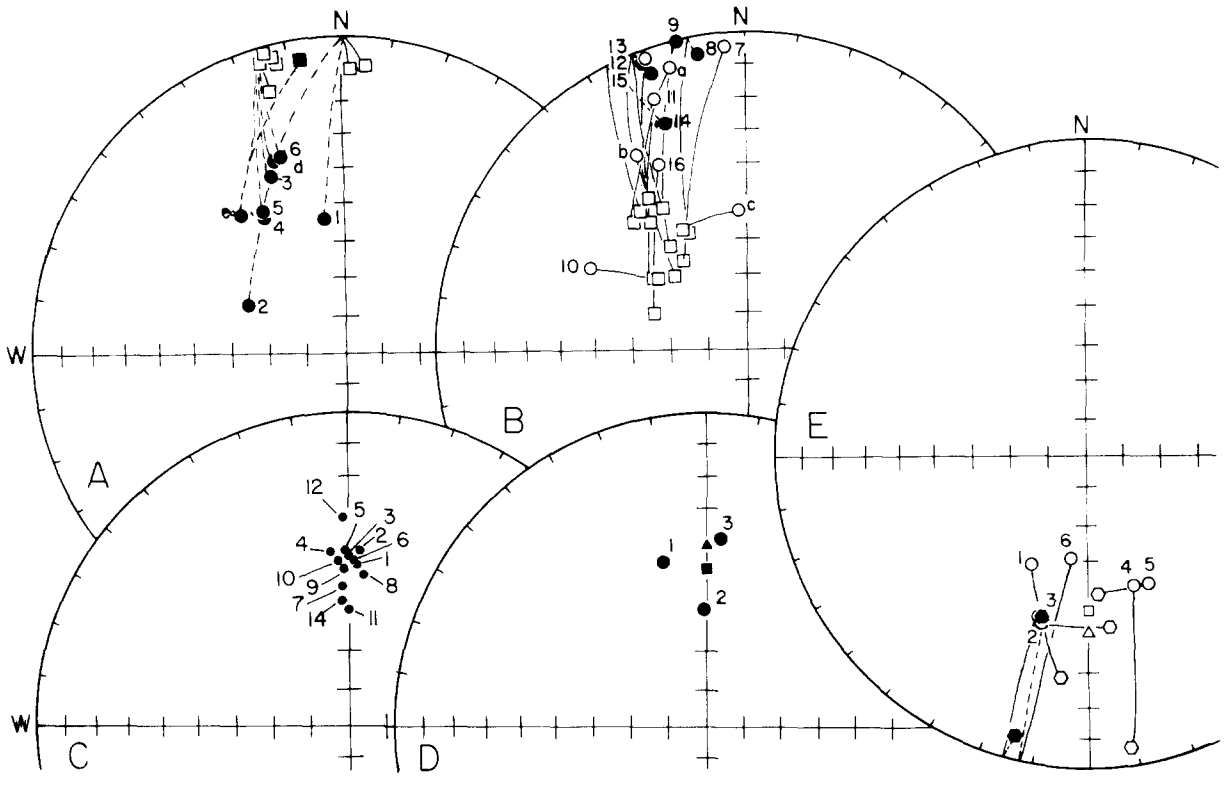


Fig. 6. Mean directions of magnetic components obtained from the various sampled successions represented in equal area projection. Results *a–e* (A, B) are of lower accuracy ($\alpha_{95} > 10°$) and have not been considered in the comparison with the Indian directional pattern (Fig. 9). Full (open) symbols represent lower (upper) hemisphere. In A and B, circles and squares indicate directions before and after bedding correction respectively. Normal polarity directions only are shown in A–D (see Table 2).
A. Collision component: *1* = Kagbeni Sst. (NKGW), *2* = Thinigaon Lst. (NJUT), *3* = Jomoson Lst. (NJLI), *4* = Kagbeni Sst. (NKWE), *5* = Dzong Sst. (NKGV), *6* = Jomosom Qzt. (NJRH), *d* = Thini Chu Fm. (NTPA), *e* = Lumachelle Fm. (NJDO).
B. Primary component: *7* = Dzong Sst. (NKGV), *8* = Kagbeni Sst. (NKGW), *9* = Kagbeni Sst. (NKWE), *10* = Lumachelle Fm. (NDBA), *11* = Jomosom Lst. (NCJL), *12* = Jomosom Lst. (NJLI), *13* = Jomosom Qzt. (NJRH), *14* = Thinigaon Lst. (NJUT), *15* = Thinigaon Lst. (NTTR), *16* = Thini Chu Fm. (NTPA), *a* = Lumachelle Fm. (NJDO), *b* = Lumachelle Fm. (NCDL), *c* = Jomosom Lst. (NJJL).
C. Recent field component of normal polarity and characterized by a low blocking temperature spectrum. Numbers represent sampled successions as shown in Fig. 2 and as described in the text. See D for direction of the ambient field (triangle) and of the axial geocentric dipole field (square). *1* = Dzong Sst. (NKGV), *2* = Kagbeni Sst. (NKGW), *3* = Kagbeni Sst. (NKWE), *4* = Lumachelle Fm. (NJDO), *5* = Lumachelle Fm. (NCDL), *6* = Lumachelle Fm. (NDBA). *7* = Jomosom Lst. (NJJL), *8* = Jomosom Lst. (NCJL), *9* = Jomosom Lst. (NJLI), *10* = Jomosom Qzt. (NJRH), *11* = Thinigaon Lst. (NJUT), *12* = Thinigaon Lst. (NTTR), *14* = Tilicho Pass Fm.–Ice Lake Fm. (NSCD). Not corrected for bedding.
D. Recent field component of normal polarity and characterized by high blocking temperature spectra: *1* = Jomosom Qzt. (NJRH), *2* = Thinigaon Lst (NJUT), *3* = Thinigaon Lst. (NTTR). Triangle = ambient field direction, square = axial geocentric dipole field direction. Not corrected for bedding.
E. Recent field component of reversed polarity and characterized by intermediate blocking temperature spectra: *1* = Lumachelle Fm. (NCDL), *2* = Lumachelle Fm. (NDBA), *3* = Jomosom Lst. (NCJL), *4* = Jomosom Lst. (NJLI), *5* = Jomosom Lst. (NJJL), *6* = Thinigaon Lst. (NJUT). Circles and hexagons indicate directions before and after bedding correction respectively.

The mean directions for the normal and reversed polarity secondary component (D), so dominantly present in the Devono-Carboniferous succession sampled at Shokang (Fig. 2, location 14), are shown in Fig. 8. MCT-related metamorphism of southwards increasing grade characterizes the southwards adjacent Nilgiri massif, but is not of penetrative nature at the sampled locality. Both the normal and the

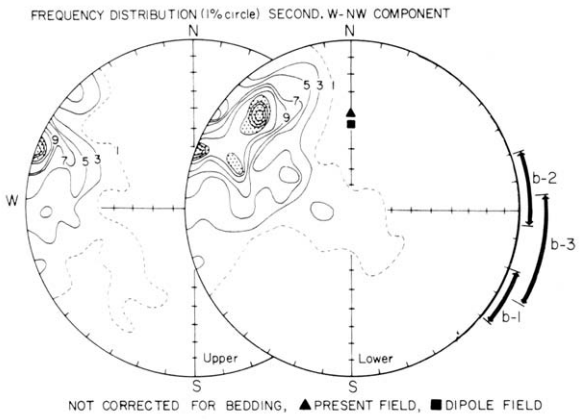


Fig. 7. Frequency distribution of all observed directions of the west to northwest component. Counted with a 1% circle. Frequency interval = 2 directions. *b-1*, *b-2* and *b-3* indicate the spread in *b*-axes of the D1-D3 tectonic phases observed throughout the Tibetan Sedimentary Series after Colchen [56].

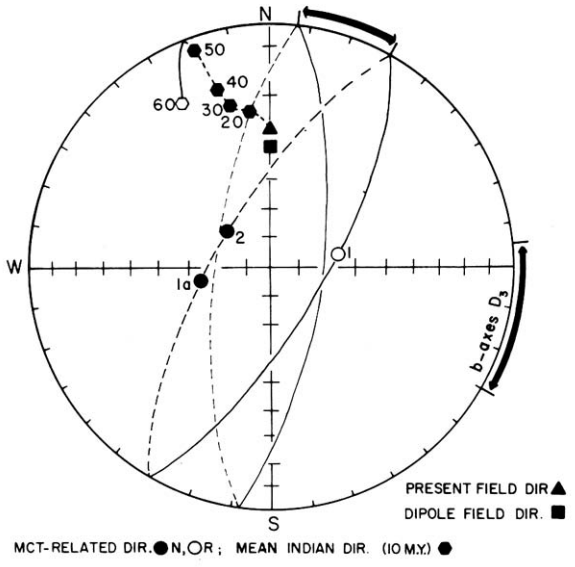


Fig. 8. Mean directions of secondary components (*1*, *1a*, *2*, not corrected for bedding) observed in the Tilicho Pass Formation–Ice Lake Formation (Fig. 2, location 14: NSCD) which are attributed to metamorphism of the basal Tibetan Sedimentary Series and of the underlying Tibetan slab during formation of and possibly during underthrusting along the MCT. *1* = reversed polarity component (*1a* = swapped antiparallel direction of *1*), *2* = normal polarity component. The hexagons represent expected directions at the sampled region according to Tertiary (DSDP core) data from the Indian plate, compiled over 10-m.y. intervals (20–60 m.y.). D-3 indicates the range of *b*-axes observed in the Tibetan Sedimentary Series for the D-3 deformation [56]. The normal and reversed polarity directions are concurrently present in nearly all specimens and have therefore a similar tectonic history. It is thus to be expected that the great circle through both directions passes through the direction of the ambient field at the time of their acquisition, and that the pole to this great circle represents the *b*-axis of the subsequent deformation. This deformation occurred during a not well-constrained period within the Miocene. Taking the latter at about 20 m.y., there is an about 20° azimuthal difference in orientation between the expected great circle through the Indian pattern (hexagons) and the actually constructed great circle. This discrepancy supports the concept of an excess counterclockwise rotation (i.e. rotational underthrusting) of India with respect to its former leading edge as discussed in the text.

reversed polarity directions come in close agreement with the late Tertiary Indian directional pattern after correction for northvergent D-3 folding, which postdates metamorphism [56].

8. Comparison with the Indian apparent polar wander path

Directions and pole positions for the collision component and the primary components are consistently displaced with respect to the expected Indian pattern (Fig. 9 and Fig. 10 respectively). As detailed in Fig. 9, the directions define a trajectory with declinations from about 10° (collision component) to about 15° (primary components) greater than on the corresponding Indian trajectory. There is no apparent inconsistency between the inclinations of the primary components and those of the Indian pattern. A similar comparison of inclinations for the collision component with the Indian pattern is not meaningful because of the nature of their dating.

The non-exact equivalence in age between individual results on both trajectories prohibits a test on statistical significance of this displacement. However, the two single results which are best comparable and which happen to be also the most accurate palaeomagnetic and best-dated results, i.e. the Rajmahal Traps (Aptian) on the Indian trajectory and the Dzong Sandstone (Lower Albian) from the Thakkhola region, are significantly different at the 95% level.

We regard it most unlikely that this consistent dis-

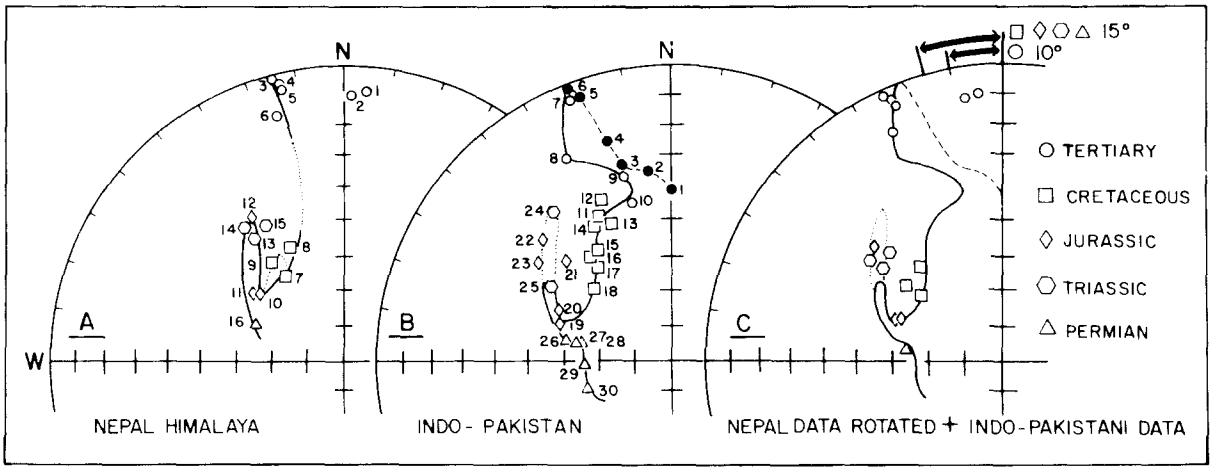


Fig. 9. Comparison of palaeomagnetic directions observed for the Thakkhola region (A) with directions expected for this region (B) according to palaeomagnetic data from the Indian subcontinent [26,28,29,71], under the hypothetical assumption that no relative movement between the two regions has occurred. The observed and the expected directions agree very well after correction for a 10° ("collision" components: *1–6*) to a 15° (primary components: *7–16*) differential rotation between the two regions (C), A. Secondary "collision" components (*1–6*; 50–60 m.y.) and primary components (*7–16*; Lower Aptian to Middle? Permian). See caption to Fig. 6A, B for description of component directions (*1–16*). B. *1* = present field direction, *2* = 5–22.5 m.y. (DSDP), *3* = 25–35 m.y. (DSDP), *4* = 35–45 m.y. (DSDP), *5* = 45–55 m.y. (DSDP), *6* = Sanjawi Limestone (Palaeocene/Eocene, *7* = Upper Brewery Limestone (base Upper Palaeocene), *8* = 55–65 m.y. (DSDP), *9* = Upper Deccan Traps normal polarity (60–65 m.y.), *10* = Lower Deccan Traps reversed polarity (60–65 m.y.), *11, 12* = Tirupati beds (Early Cretaceous), *13* = 65–75 m.y. (DSDP), *14* = Satyavedu beds (Early Cretaceous), *15* = Sylhet Traps (~Rajmahal Traps), *16, 17* = Goru Formation–Parh Limestone (Aptian-Albian to Santonian-Campanian), *18* = Rajmahal Traps (100–105 m.y.), *19* = Loralai Limestone (Middle/Upper Jurassic), *20* = Chiltan Limestone (Upper Jurassic), *21* = mean Middle Jurassic pole for Australia transferred to the Indian plate [29], within the adapted Smith and Hallam reconstruction of Gondwanaland [96–98], *22* = mean Lower Jurassic pole for Australia [29], transferred as above, *23* = mean Middle Jurassic pole for Antarctica [29], transferred as above, *24* = Parsora beds (Upper Triassic), *25* = Pachmarhi beds (Upper Triassic), *26* = Kamthi beds, Wardha Valley (Upper Permian/Lower Triassic [57]), *27* = Mangli beds (Upper Permian/Lower Triassic), *28* = Panchet beds (Upper Permian/Lower Triassic), *29* = Kamthi beds, Wardha Valley (Upper Permian /Lower Triassic [99]), *30* = Talchir beds (Permo-Carboniferous).

placement is an artifact resulting from our interpretation of the various magnetic components, as both the Indian and the Thakkhola trajectory are similar in shape but not in position and show high resolution in time, in particular the Indian Tertiary segment. Consequently, we interpret the 10–15° difference in declination between the expected and the observed directional pattern as approximate lower and upper limits for the magnitude of a differential rotational movement since early Tertiary times between India and the TSS of the Thakkhola region, or possibly in a wider perspective between India and its former leading edge as will be discussed in the next section.

These limits need further refinement and their determination warrants some further discussion:

(1) We have ignored in our comparison low accuracy results for the primary components (Fig. 6B, *a, b, c*) and for the collision component (Fig. 6A: *d, e*), i.e. results with α_{95} values in excess of 10° and low N'/N ratios (Table 2), though these results tend to support the interpretation reached above. We discarded also two collision component directions (Figs. 6A, 9A, 10B: *1, 2*) which are distinctly off the main grouping. We can offer no explanation for their offset, other than pointing to the possibility of an undetected tectonic complication. Inclusion of both directions tends to equate the lower limit value with the upper limit value, but we see no justification for such a procedure.

(2) The declination pattern of the Tertiary part of the Indian trajectory (past 60 m.y.) is not based on direct palaeomagnetic observations, but on a compilation of pole positions determined from palaeolatitude observations on DSDP cores, in combination with sea-

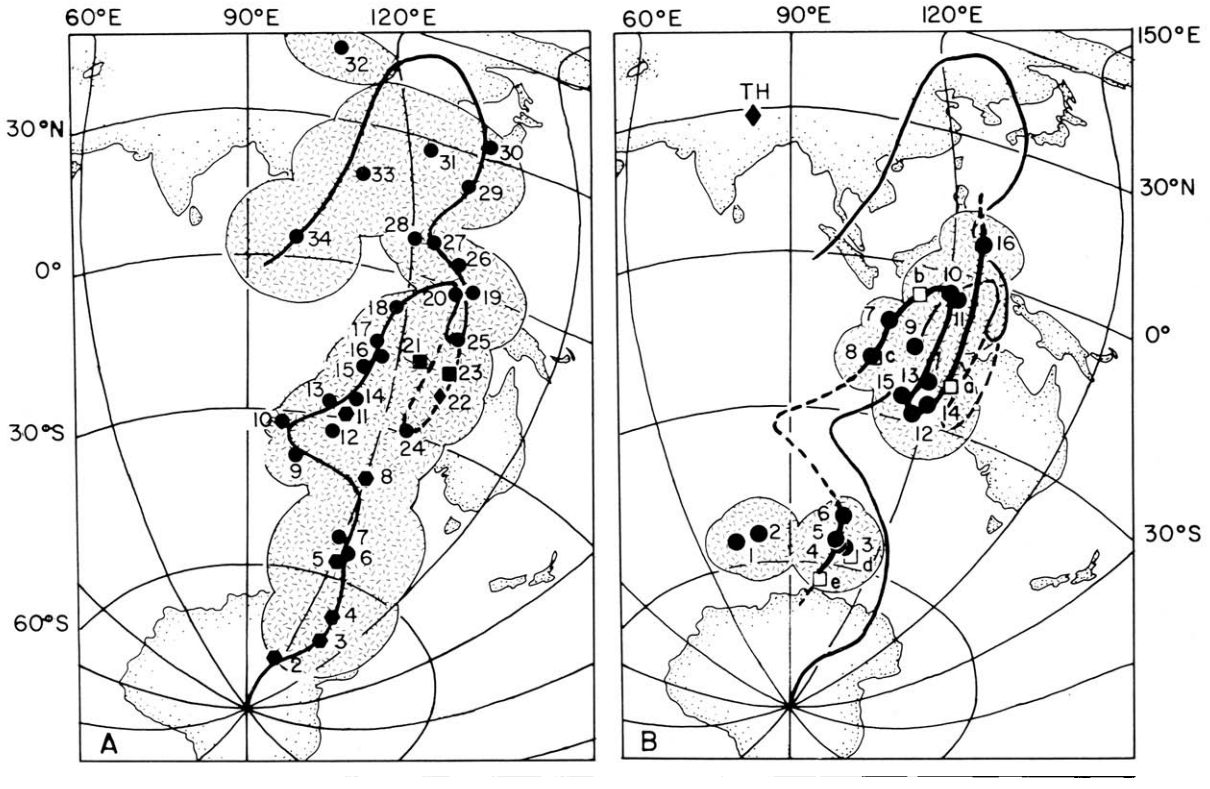


Fig. 10. Comparison of the Indian APWP (A) with pole positions corresponding to the primary and secondary collision components (B). The eastwards displacement of the latter with respect to the former reflects rotational underthrusting of India beneath its former leading edge. Legend: for Fig. 10A, see legend to Fig. 9B for poles 1–30, *31* = Krol-A Limestone (Permo-Triassic), *32* = Blaini Limestone (Permo-Carboniferous), *33* = Alozai Formation (Permo-Carboniferous), *34* = Lower Blaini Diamictite (Permo-Carboniferous). For Fig. 10B, see legend to Fig. 6A, B and Table 2.

floor spreading data [26,50,73]. We regard the accuracy of this data set to be at least equal to any forthcoming directly determined Tertiary palaeomagnetic data for the Indian subcontinent, but await such data for further confirmation.

(3) Directions and pole positions for the primary components (Middle? Permian to Early Cretaceous) show a distinct early Mesozoic loop (Figs. 9A, 10B: *16-10*), which can be identified also in the Indian pattern (Figs. 9B, 10A: *28-19*), though of less clearly defined extent. One of us (C.T.K.; see also [26,74]) previously attributed apparently off-track Permian and Triassic results from Indian Gondwana beds to an early Tertiary magnetic overprint associated with the Deccan Trap magmatism. A renewed study of Indian Gondwana beds (Agarwal and Klootwijk, in progress) shows further evidence for an early Tertiary magnetic overprint of in particular Permian Gondwana beds. Importantly, this reappraisal also shows Middle- and Late Triassic directional results which support the reality of this early Mesozoic loop (i.e. positioned between datapoints *24* and *25* in Figs. 9B and 10A). Further support for the reality of this loop comes from Middle to Late Jurassic results from Baluchistan [29] and from Early and Middle Jurassic mean poles for Australia and Antarctica, transferred to India according to seafloor spreading data (Figs. 9B, 10A) [29].

(4) The normal and reversed polarity directions of the MCT-related secondary component observed at locality 14 (Fig. 2, Table 2) can be interpreted in further support of the suggested differential rota-

tional movement. The precisely determined normal and reversed polarity directions are significantly non-antiparallel at the 95% level (Table 1, Fig. 8). Both results are obtained from the same samples, therefore this discrepancy has to be explained by acquisition of these components at different stages of the D-3 deformation phase [56]; with the reversed component (Fig. 8: *1, 1a*) acquired prior to the normal component. Under this assumption the great circle through both components should pass through the ambient field direction at the time of their acquisition with the pole to the great circle coinciding with the *b*-3 axis of folding, as this pole indeed does (Fig. 8) [56, p. 98]. This great circle is, however, about 20° clockwise displaced from the late Tertiary (about 20 m.y.) Indian direction. We interpret this mismatch as further support for the above detailed sense and magnitude of differential rotational movement.

9. Magnitude of underthrusting along the Main Central Thrust

The consistently deviatory pattern of declinations for the Thakkhola region is based on data from three different structural units [56] within the TSS which are well continuable along strike into regions adjacent to the Thakkhola graben. We therefore interpret this consistent mismatch between observed and expected declinations to represent a differential rotational movement along the MCT of India with respect to the central Nepal segment of India's former leading edge. We do not favour the interpretation that this mismatch reflects local rotations within the TSS of the Thakkhola region. It remains open to further study to establish whether the 10–15° mismatch may be representative for a wider part of India's former leading edge. This may well prove to be so, notwithstanding more complex movements in the northwestern part of this former leading edge [30], as the surface expression of the MCT and the structural trends of the TSS are well continuable throughout the central Himalaya. It is therefore a tempting speculation to define from this 10–15° differential rotation palaeomagnetic constraints on the magnitude of rotational underthrusting along the MCT.

The counterclockwise rotational northward movement of the Indian plate [25,35,36] and indications for an eastward increasing magnitude of underthrusting along the MCT [25,37], suggest that the pole for this rotational underthrusting has to be sought for in the northwestern part of the Indian subcontinent [37]. As a first approximation we assume its location at the western end of the Indus Kohistan Seismic Zone (Fig. 1) [75] which though of an apparently different tectonic regime, represents the subsurface continuation of the MCT-MBT complex beyond the Western Himalayan Syntaxis. Both the MCT and the MBT follow more or less the same tectonic belt in the region directly east of the Western Himalayan Syntaxis (Fig. 1), but are distinctly separate entities further to the east.

Intracontinental underthrusting along the MBT has been suggested [49] but is disputed [2] as discussed before. Relative movement along the MBT seems to be more of translational than of rotational character and seems of minor magnitude compared with movement along the MCT. As a further approximation we assume therefore that rotational underthrusting occurred along the MCT only. Under these assumptions the 10–15° differential rotation determined here would correspond to a minimum underthrusting of India beneath its leading edge over 200–350 km at the longitude of central Nepal. This interpretation is not in disagreement with recent model studies of bouyance constraints on subduction of continental lithosphere [76].

There is consensus from geological data ([49], P. Le Fort, personal communication, 1980), deep seismic sounding data [77], gravity data [78,79] and fault plane solutions [80,81] for an about 15° northward dip of the MCT. Assuming this dip to be uniform, the corresponding surface projection of the frontal part of underthrust India is shown in Fig. 1 by the two heavy lines which represent the lower (10°) and the upper (15°) limit on the magnitude of the rotational underthrusting. Depths to the top of the underthrusted slab at the "15°" line range from about 60 km north of Mount Kailas to about 120 km north of Lhasa.

We have previously suggested [32] a relation between this underthrusted slab and the calc-alkaline volcanism in southern Tibet (Fig. 1), assuming a Neogene age for the latter [82,83]. Newly available radiometric data suggest, however, an early Tertiary or older age for these volcanics (A. Gansser, personal

communication, 1980 – 40 m.y., Rb-Sr whole rock isochron; Zhang Yuquan, personal communication, 1980 – 50 m.y. or older; Jin Chengwei [46] – Cretaceous or early Tertiary). Their extrusion thus preceded the late Tertiary underthrusting along the MCT, and the coincidence between the surface projection of the frontal part of the subducted continental slab and the calc-alkaline volcanism (Fig. 1), is therefore purely fortuitous.

10. Northern extent of India and other implications

The above constraints on intracontinental underthrusting along the MCT in combination with earlier estimates of crustal shortening within the Himalayan belt allow a minimum estimate of India's northern extent (post-Neotethys formation), shown in Fig. 11.

Our estimate is based on: (a) 200 km crustal shortening through telescoping between the MBT and the ITS, excluding underthrusting along the MCT; (b) 50 km subduction of continental lithosphere along

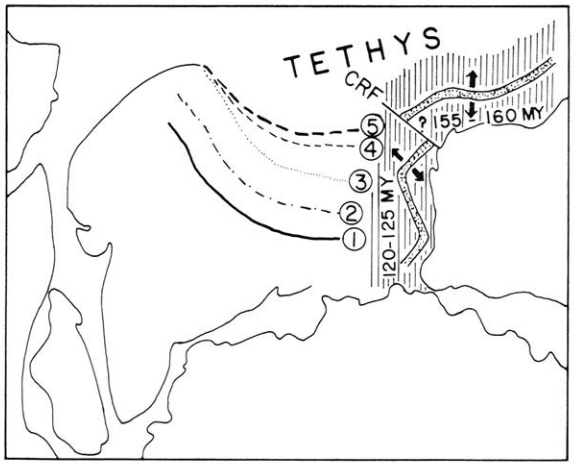


Fig. 11. The northern extent of India within the Smith and Hallam [96] reconstruction as adapted after Griffiths [97] and Norton and Molnar [98]. *1–5* are five lines relevant to determining India's northern extent as discussed in the text. *1* = the present Himalayan front (MBT), *2* = the Indus-Tsangpo suture zone (ITS), *3* = a reflection of the ITS and is separated from *2* by a 300-km-wide belt representing: (a) 200-km crustal shortening within the Himalayan region except movement along the MCT, (b) 50-km continental subduction along the ITS, and (c) 50-km northward translational underthrusting along the MCT. The zone between line *3* and line *4* or *5* represents the contribution to India corresponding with rotational underthrusting along the MCT over 10° or 15°, respectively. Rift zones (stippled) and assumed continental lithosphere (vertical shading) according to Veevers and Cotterill [87]. Ages of commencement of spreading are indicated. The magnitude of the age discontinuity along the Cape Range fracture zone (CRF) and the continuation of the Jurassic rift zone to the northwest of the Exmouth Plateau are disputed [87,88].

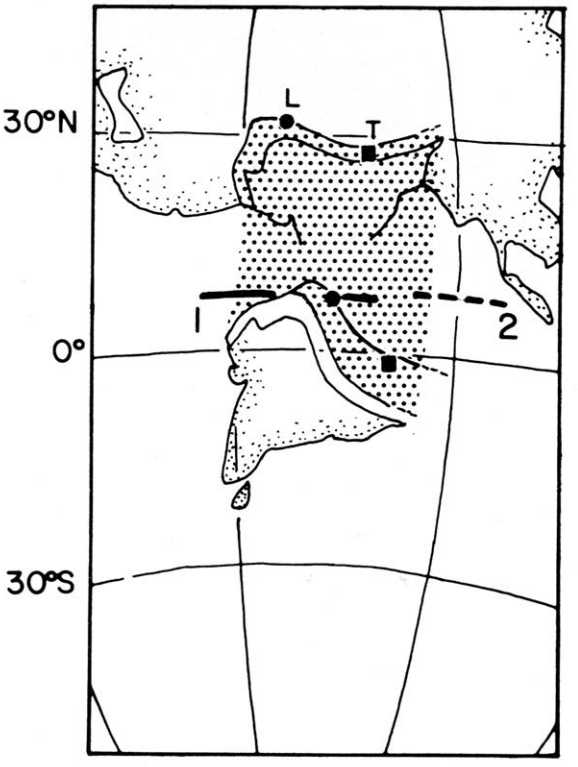


Fig. 12. Comparison of India's and Asia's palaeolatitudinal positions at the time of initial collision (about 55 m.y.). India is positioned according to the here detailed secondary collision component (Fig. 6A: *3–6*) from the Tibetan Sedimentary Series of the Thakkhola region (location indicated by square *T*). Asia is positioned according to Irving's [27] mean pole position for Eurasia (55 m.y.). Its southern boundary is shown under assumption of no post-collisional deformation within south central Asia. The so obtained palaeoposition of India upon collision is in good agreement with the contemporaneous palaeoposition of the Ladakh Intrusives (heavy line through dot [28]), which are now situated directly north of the ITS (present position indicated by dot *L*). The dashed line *2* represents the palaeoposition obtained from Upper Cretaceous/Lower Tertiary red beds in southern Tibet, north of Lhasa [93.94]. The stippled zone indicates the magnitude of post-collisional crustal shortening within India and southern Asia.

the ITS; (c) a similar estimate for northward translational underthrusting along the MCT; and (d) rotational underthrusting along the MCT over 10–15°.

The values under (a) and (b) are conservative quotes of generally accepted magnitudes [23,25,48, 49,79,80,84,85]. The magnitudes under (a), (b) and (c) combined account for line *3* in Fig. 11, which reflects the ITS at a uniform distance 300 km to the north. The 10–15° estimates for rotational underthrusting account for line *4* and line *5*, respectively, which represent our minimum estimates for India's northern extent. In particular the 15° estimate (line *5*, Fig. 11) well satisfies constraints from seafloor spreading data of the eastern Indian Ocean and facies analyses off western Australia's margin [86–89]. Veevers and Cotterill's [87] study interprets reflection profiles and drilling data in terms of the former existence of an extensive continental lithospheric block to the west of the western Australian margin at least as far north as the Cape Range fracture zone border of the Exmouth Plateau. A more limited strip of continental lithosphere may have existed off northwestern Australia. Seafloor spreading data indicate that rifting off northwestern Australia started at about 155–160 m.y. ago [88,90], whereas the continental lithospheric block off western Australia drifted away from about 120 to 125 m.y. onwards [87–92].

Clearly, our preferred model (Fig. 11, line *5*) falls far short of Powell and Conaghan's original "Greater" India model [2,3,31] which figured a postulated underthrusting of "Greater" India beneath the full width of the Tibetan Plateau. Our model is, however, comparable with Powell's [25] revised model and Johnson et al's [89] model.

India's palaeolatitudinal position at the time of acquisition of the secondary collision component is shown in Fig. 12. The palaeoposition of the juxtaposed southern boundary of central Asia is not well determined as yet and is constrained only by palaeomagnetic data from the Ladakh Intrusives [28] located directly north of the ITS (Fig. 12, full dot *L*), and by Late Cretaceous/early Tertiary red bed data from southern Tibet, north of Lhasa [93,94]. The palaeolatitude of the Ladakh Intrusives at the time of collision (7–10°N [28]; Fig. 12, heavy line *1* through full dot) is in very good agreement with the expected palaeoposition of India's northern boundary in the Ladakh region according to the secondary collision component reported here. The Tibetan red bed data indicate a palaeolatitude of 8°N for the sampled area in southern Tibet (Fig. 12, dashed line *2*). It is unfortunate that poor age control reduces their usefulness for further delineation of central Asia's southern boundary at the time of collision.

Somewhat oversimplifying the collision process, we envisage the India-Asia collision to have been initiated along a zone represented in position by India's northern boundary as sketched in Fig. 12. The equatorial to low northern palaeoposition of this collision zone is in distinct mismatch with its expected location at about 30°N according to Eurasian palaeomagnetic data ([27]: 55 m.y.), assuming no post-collisional deformation within south central Asia. This mismatch over 2500–3000 km can possibly be accounted for by effects of (a) closure of small ocean-floored basins, (b) crustal thickening through basement reactivation within the Pamir knot, the Tibetan Plateau and the mountain belts to the north [22], or (c) lateral movement of lithospheric blocks within Asia out of the way of the advancing Indo-Pakistan subcontinent [23].

The contribution from each of these effects has not been established as yet from direct observation. There are, however, constraints on the magnitude of the first two contributions on the basis of which a minimum estimate for the third effect may be inferred. Former existence of island arcs has so far been established only for the western part of the convergence zone (Kohistan-Ladakh, [41,42]), where the Shyok zone may reflect the closure of a marginal basin. The Kangdese belt further eastwards, has been interpreted variously to represent an Andean-type margin [47] or an offshore island arc [25,43–46]. The magnitudes of effects (a) and (b) are thus less well established for the more eastern part of the collision zone and we confine ourselves in the following discussion to its western part. The extreme case that the India-Asia convergence between initial contact of India with an island arc offshore Asia, at about 55 m.y. ago [3,25,28,30,73] and final jamming of the Indian and Asian continental blocks at about 40 m.y. ago [23,50] was taken up in full by closure of marginal basins, would lead to a maximal magnitude for effect (a) of about 1500 km. Crustal shortening through crustal thickening within the Pamir block (b)

has been estimated at 200–300 km [23]. This leaves a balance of more than 1000 km, which may be attributed to effect (c). This conclusion is in remarkable agreement with palaeomagnetic evidence for about a 1000 km of crustal convergence since the Palaeogene within Asia north of the Pamir-Transalai zone [30,71, 95], and supports Molnar and Tapponnier's [23] lateral displacement model.

11. Conclusions

Late Palaeozoic to early Tertiary palaeomagnetic results from the Tibetan Sedimentary Series of the Thakkhola region of central Nepal resemble Indian palaeomagnetic data, but show a consistent 10–15° offset in declination. This offset has been interpreted tentatively to represent rotational underthrusting along the Main Central Thrust of India beneath its former leading edge and the Tibetan Plateau. According to the model presented here, underthrusting of continental lithosphere may have proceeded over 200–350 km at the longitude of the Thakkhola region. A minimum estimate for India's northern extent (post-Neotethys formation) is proposed, which accounts for this magnitude of underthrusting and for crustal shortening within the Himalaya. Within the modified Smith and Hallam reconstruction of Gondwanaland, this minimum estimate well satisfies seafloor spreading data from the eastern Indian Ocean and facies constraints from the western Australian continental margin. Available palaeomagnetic data from the Himalayan region show that India's indentation of south central Asia exceeds 2500 km, which in part can be interpreted in support of Molnar and Tapponnier's continental collision model.

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